

Design of a Multipurpose Sieving Machine Power Transmission System

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PRESENTATION OVERVIEW

01. Project overview and problem statement
02. Design ideas
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04. Stress analysis
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PROBLEM STATEMENT

OBJECTIVE

Design a mechanically operated sieving machine that separates granular materials into coarse and fine categories. Operate at uniform vibration speed, handle 3-4 kg load, and perform continuous sieving with minimal effort

DESIGN REQUIREMENTS

The design should be structurally stable, easy to operate, and cost-efficient.



REAL-WORLD APPLICATION:

Sieving machines are widely used in agriculture, construction, pharmaceuticals, and food industries for separating materials like sand, grains, powders, and cement.

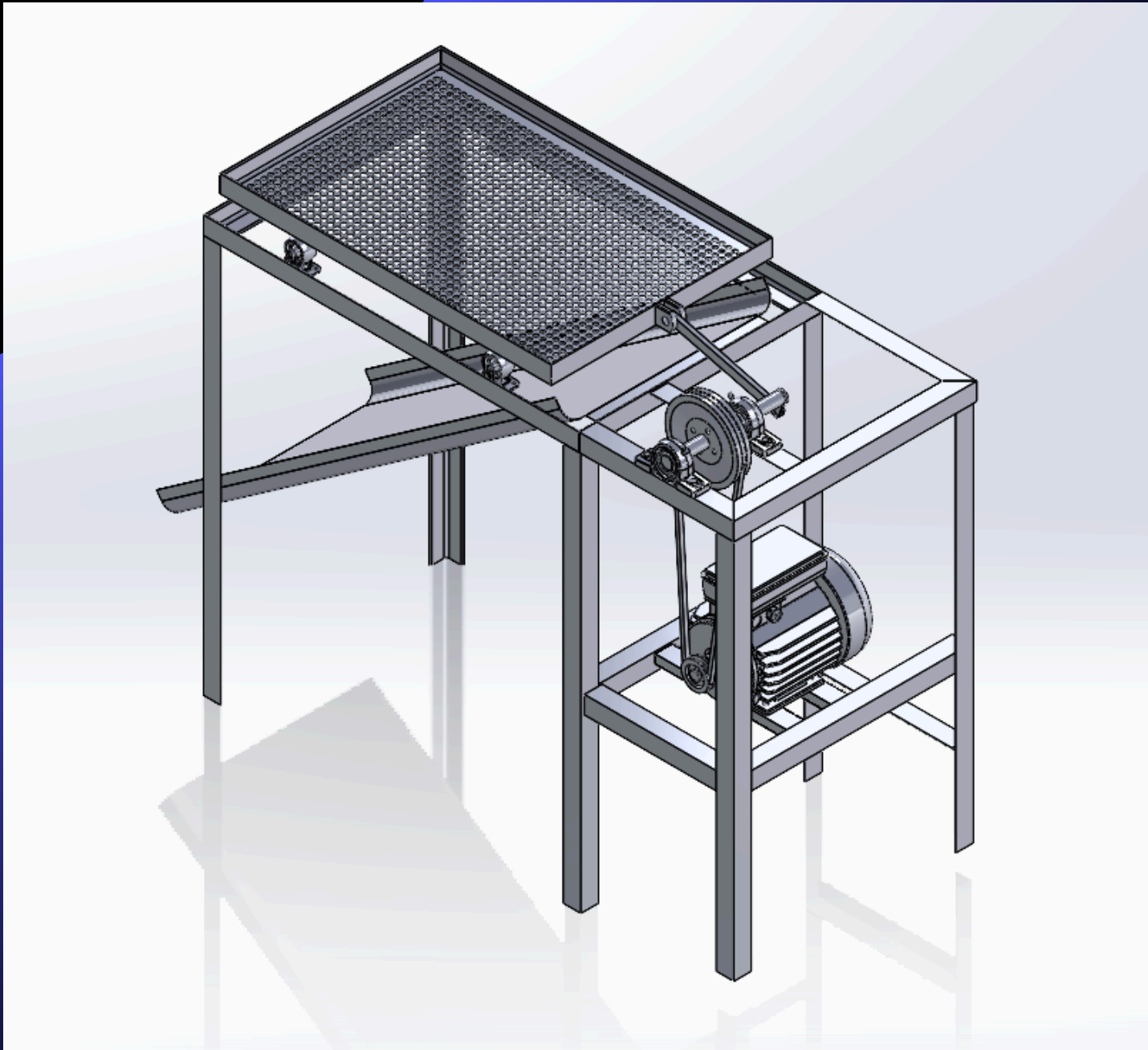
Increased Efficiency:

Manual sieving is time-consuming and inconsistent. A mechanical sieving machine improves both speed and accuracy

User Safety & Convenience:

A stable and easy-to-operate machine reduces operator fatigue and ensures consistent product quality.





Technical Learning:

The project integrates concepts from mechanical vibrations, power transmission, structural analysis, and machine design.

Cost-Effective Solution:

Most industrial vibrating sievers are expensive. Designing a compact, low-cost system helps small businesses and labs

·**Capacity Requirement:**

The machine must process 2–3 kg of mixed material per batch.

·**Separation Efficiency:**

It should effectively separate into at least two size fractions, based on mesh size.

·**Vibration Mechanism:**

A motor-driven eccentric cam or crank-slider must produce uniform vibration.

·**Structural Stability:**

The supporting frame must withstand continuous vibration without permanent deformation.

·**User-Friendly Operation:**

It should allow easy mesh replacement, material loading, and collection.

The first idea was to use a reciprocating hand-operated system similar to traditional manual sieves.

However, several issues were identified:

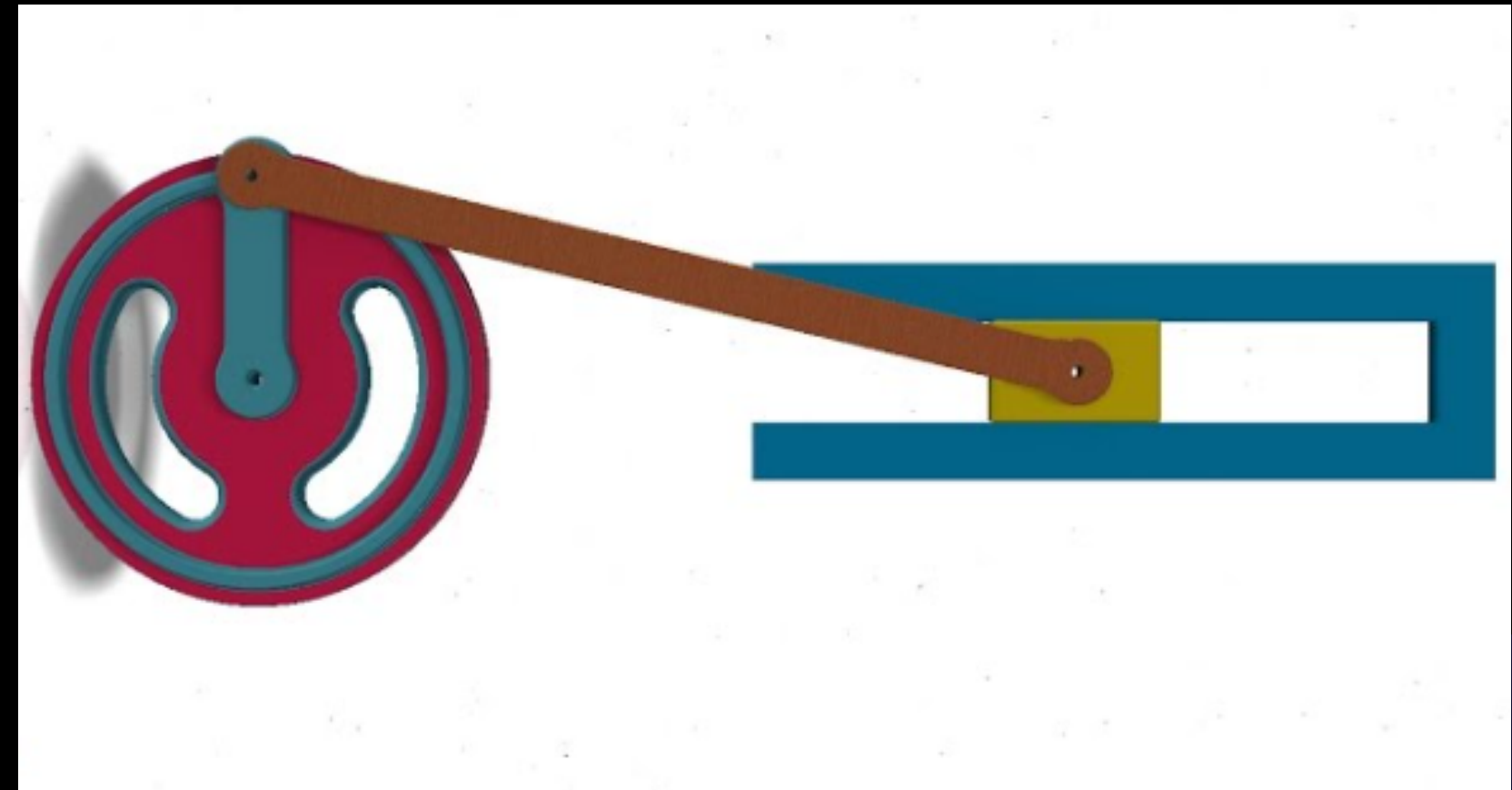
- Inconsistent vibration amplitude due to manual operation.
- Low output, not suitable for bulk separation
- Operator fatigue and varying efficiency
- Lack of structural stability for heavy or continuous usage.

Due to these limitations, the idea was rejected



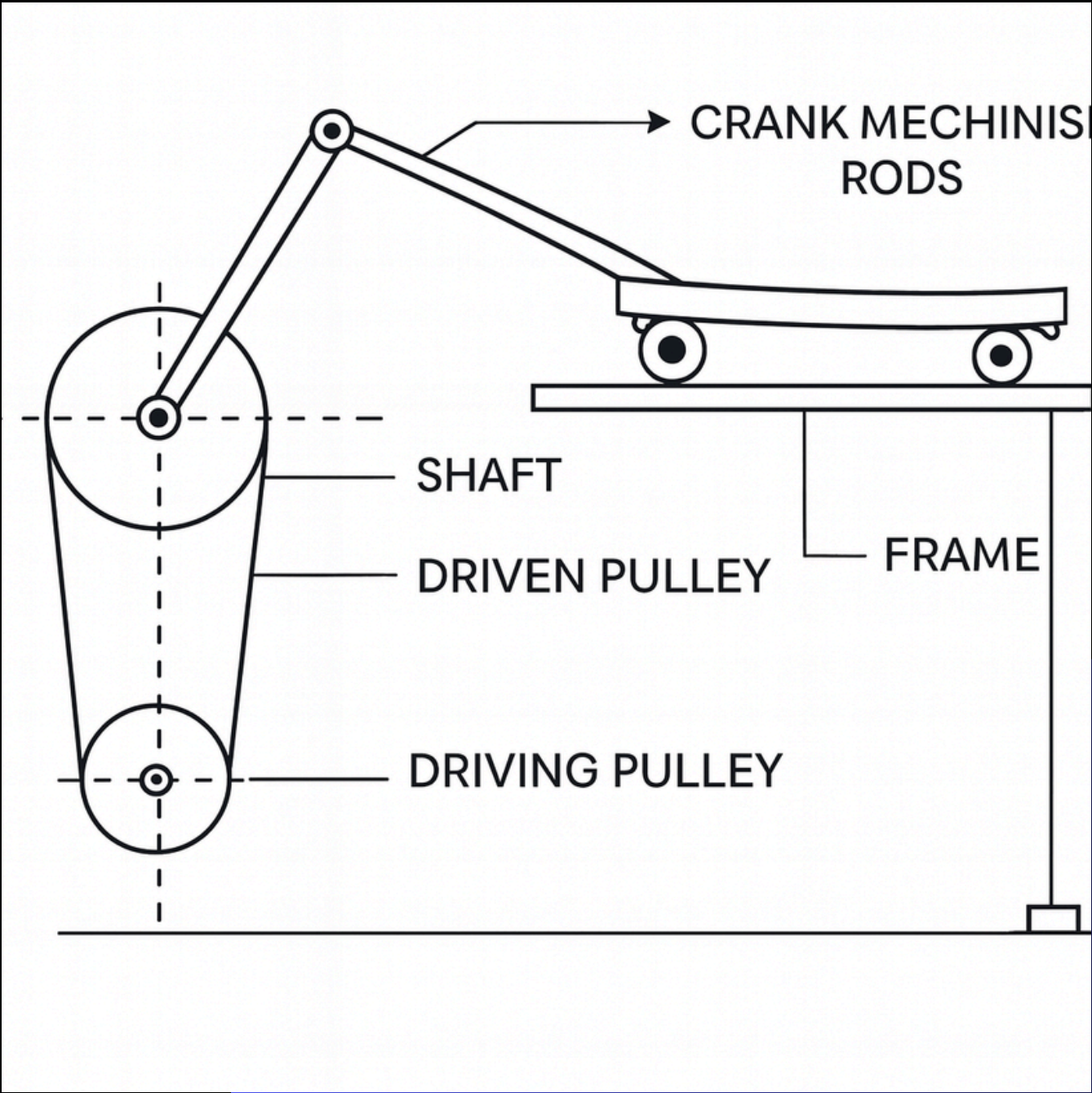
Final Solution

The final design uses a motor-driven crank–slider mechanism to generate controlled reciprocating motion for the sieve tray. The crank–slider provides a simple way to convert rotary motion of the motor into a linear reciprocating motion.



Design Calculations

Component	Mass (kg)
Driving pulley	0.158
Driven pulley	3.757
Belt	0.523
Shaft	1.046
Crank rods	0.381
Frame	5.301
Material sieved	3.000
Total Mass	14.009 kg
Radius of driving pulley: 22.5 mm	



Design Calculations

Torque Required

So basically, this weight has to be balanced by the torque of the motor.

$$T = mgr$$

$$T = 14 \times 9.8 \times 22.5 \times 10^{-3}$$

$$T = 3.087 \text{ Nm}$$

Applying Factor of Safety

Let the factor of safety of the motor be:

$$f_s = 1.5$$

$$T_{\max} = f_s \times T$$

$$T_{\max} = 1.5 \times 3.087 = 4.63 \text{ Nm}$$

Selecting Motor RPM

For this value of maximum torque, we get the required RPM from the standard motor chart.

From the chart, we get:

$$\text{RPM} = 1500 \text{ rpm}$$

Power Calculation

$$P = T \times \omega$$

$$\omega = 2\pi N/60$$

$$P = 4.63 \times \frac{2\pi \times 1500}{60}$$

$$P = 727.27 \text{ W}$$

$$\approx 1 \text{ Horsepower}$$

Design Calculations

As the required power is near 750 W, which corresponds to 1 HP, we choose a 1 HP, 1500 rpm motor.

Belt Length Calculations

We can calculate the belt length using the following data:

- D (Diameter of driven pulley) = 0.145 m
- d (Diameter of driving pulley) = 0.045 m

Speed Ratio

$$\text{Speed Ratio} = \frac{d}{D} = \frac{0.045}{0.145} = 0.31$$

Angular Velocity of Driven Shaft

$$\omega = N_{\text{driving}} \times \frac{d}{D} \times \frac{2\pi}{60}$$

$$\omega = 1500 \times 0.31 \times \frac{2\pi}{60}$$

$$\omega = 48.69 \text{ rad/s}$$

Centre Distance Between Pulleys

$$x = 0.47 \text{ m}$$

Length of Belt

$$L = \frac{\pi}{2}(D + d) + 2x + \frac{(D - d)^2}{4x}$$

Substituting values:

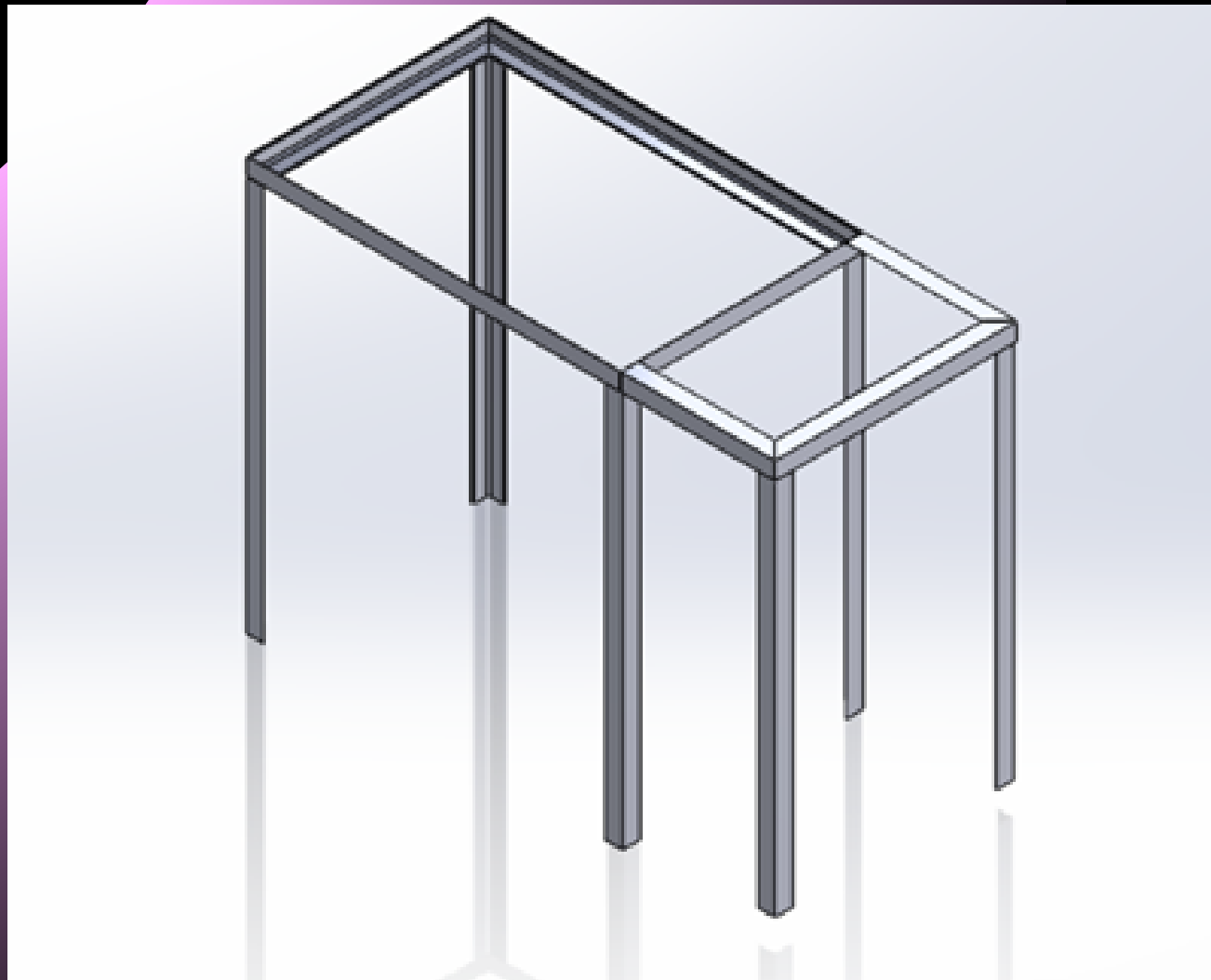
$$L = \frac{\pi}{2}(0.145 + 45 \times 10^{-3}) + 2(0.47) + \frac{(0.145 - 0.045)^2}{4 \times 0.47}$$

$$L = 1.25 \text{ m}$$

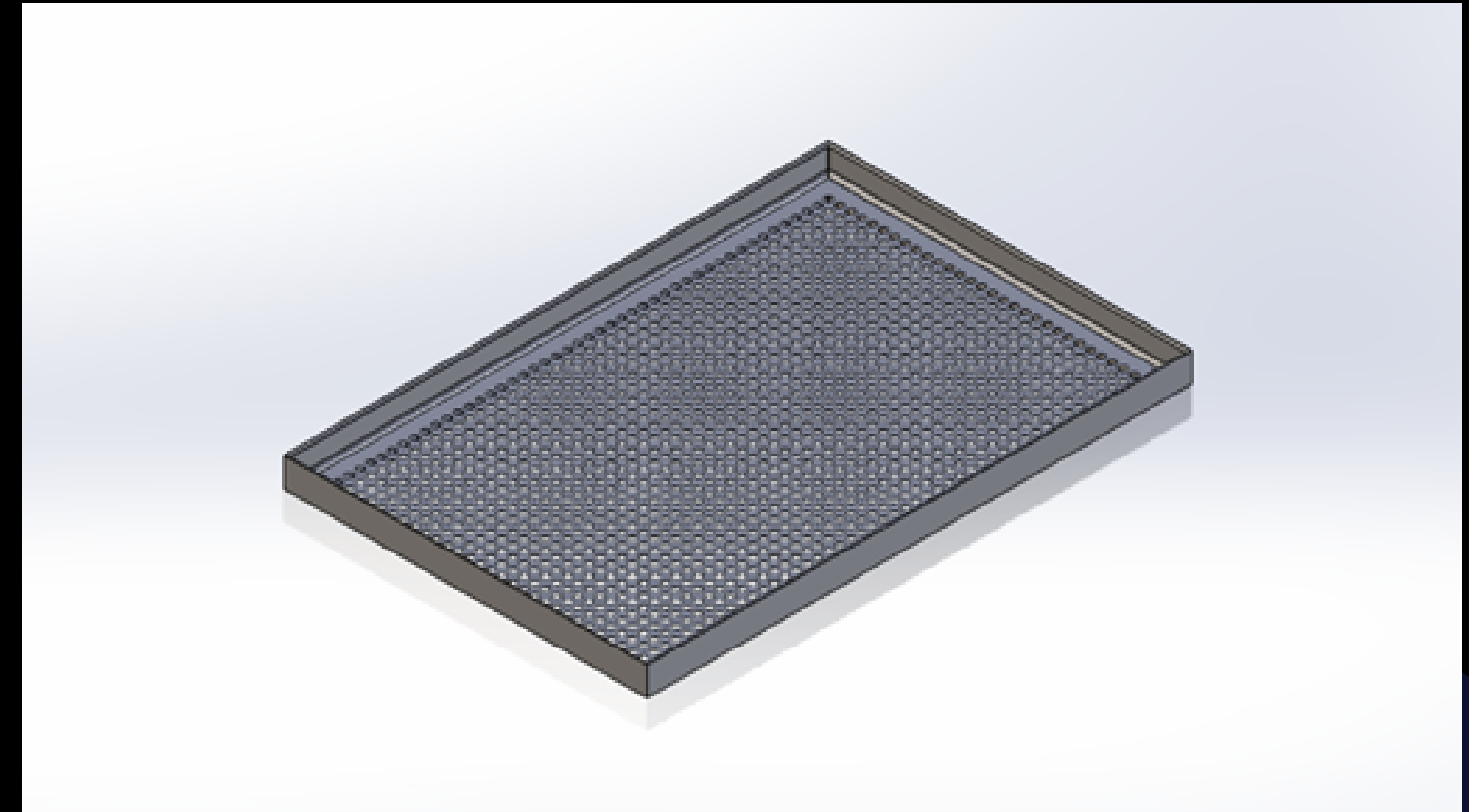
Materials

- **Pulleys:** Cast Iron / Mild Steel
- **Belt:** Neoprene rubber with polyester cords
- **Shaft:** EN8 steel
- **Crank rods:** Mild steel / EN8
- **Bearings:** CI housing + AISI 52100 balls
- **Frame:** MS square tubes (IS 2062)
- **Mesh:** Stainless steel (SS304)

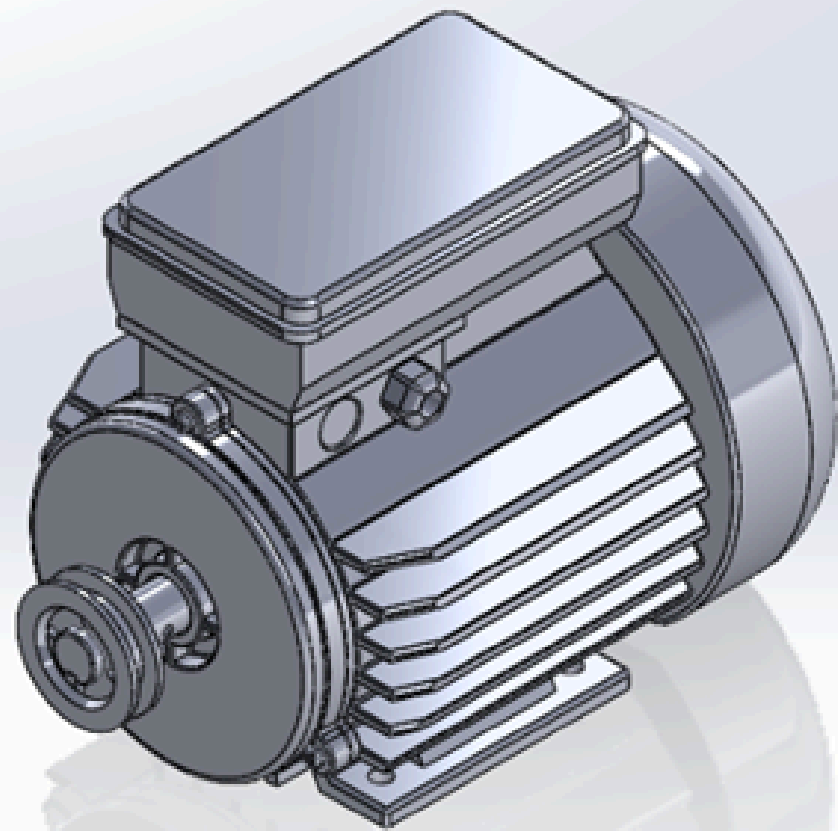
BASIC STRUCTURE



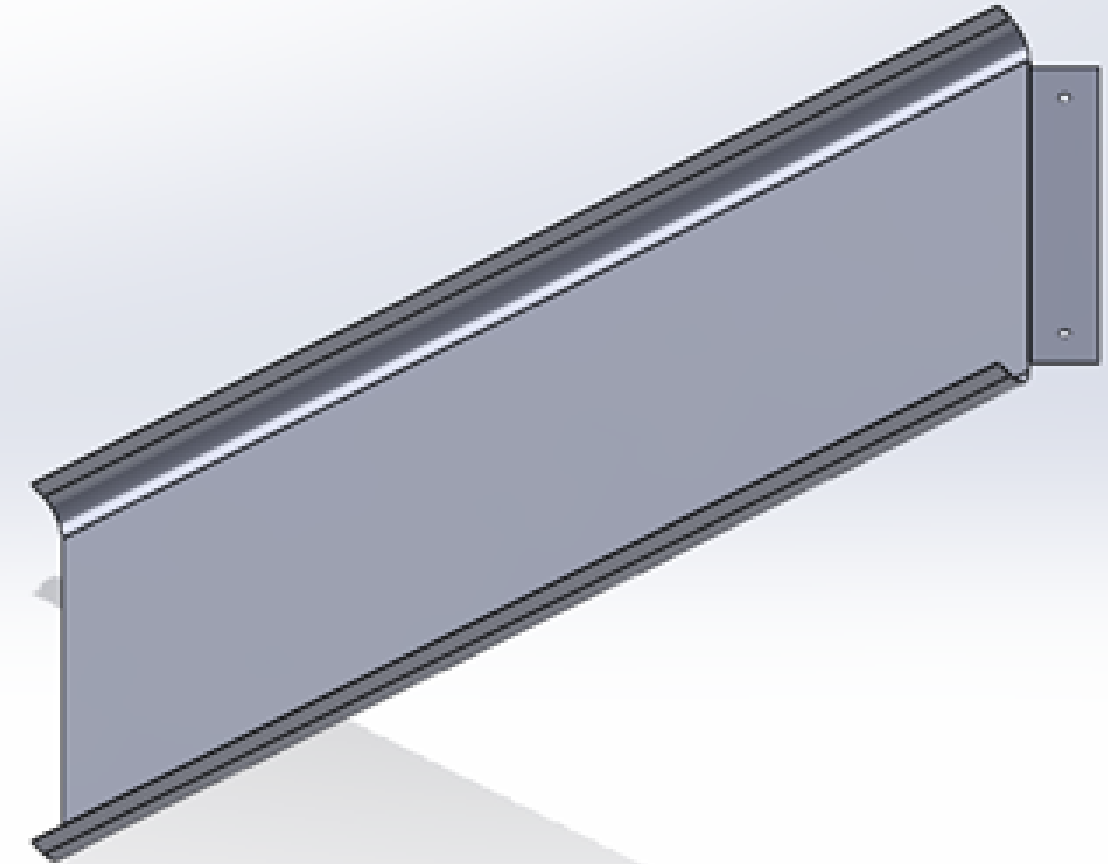
- Rectangular mild steel frame



- A sieve box with replaceable mesh plates



- A crank–slider assembly driven by a 1 HP electric motor

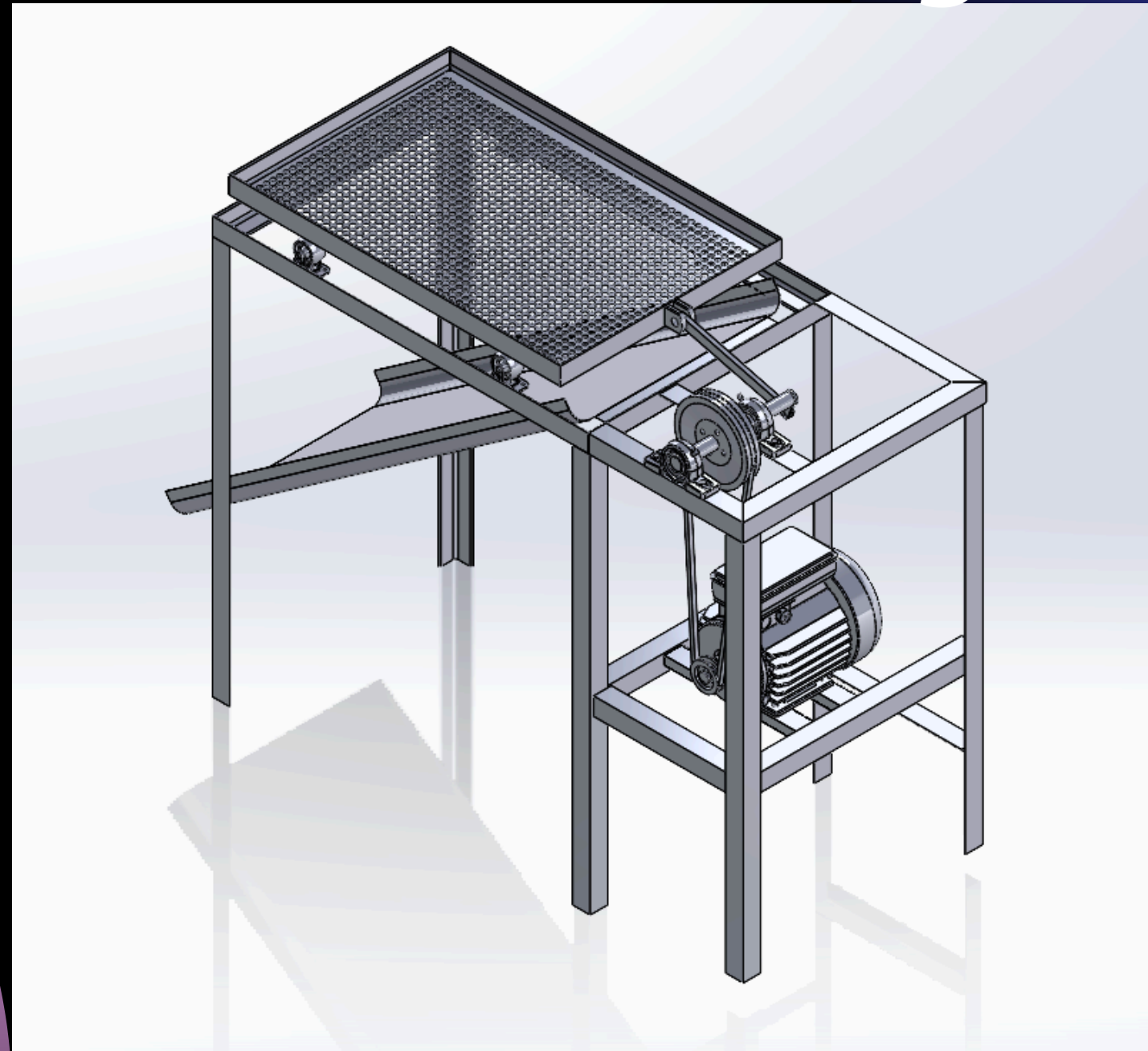


- Collecting tray below the mesh



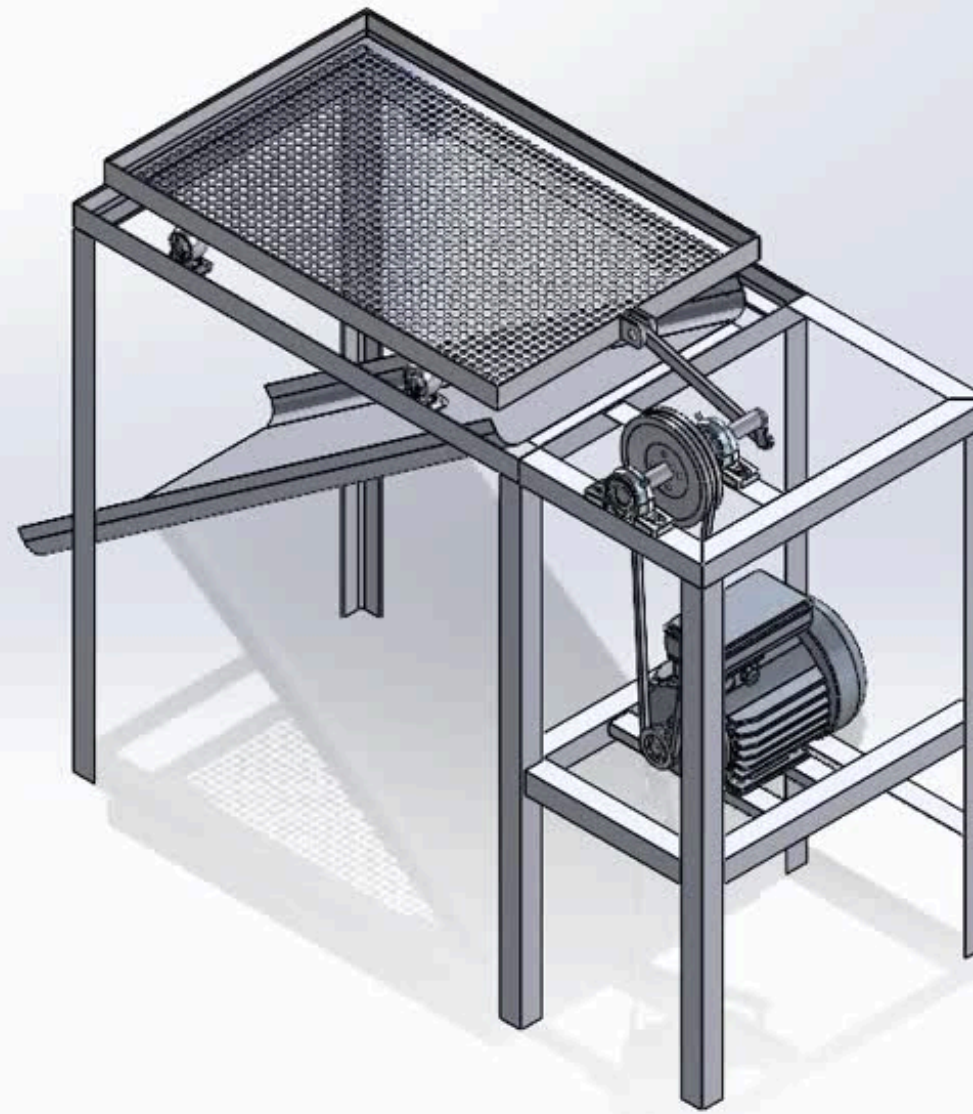
- Main shaft with a pulley supported on pillow-block bearings, connected by a belt to a lower pulley

Assembly

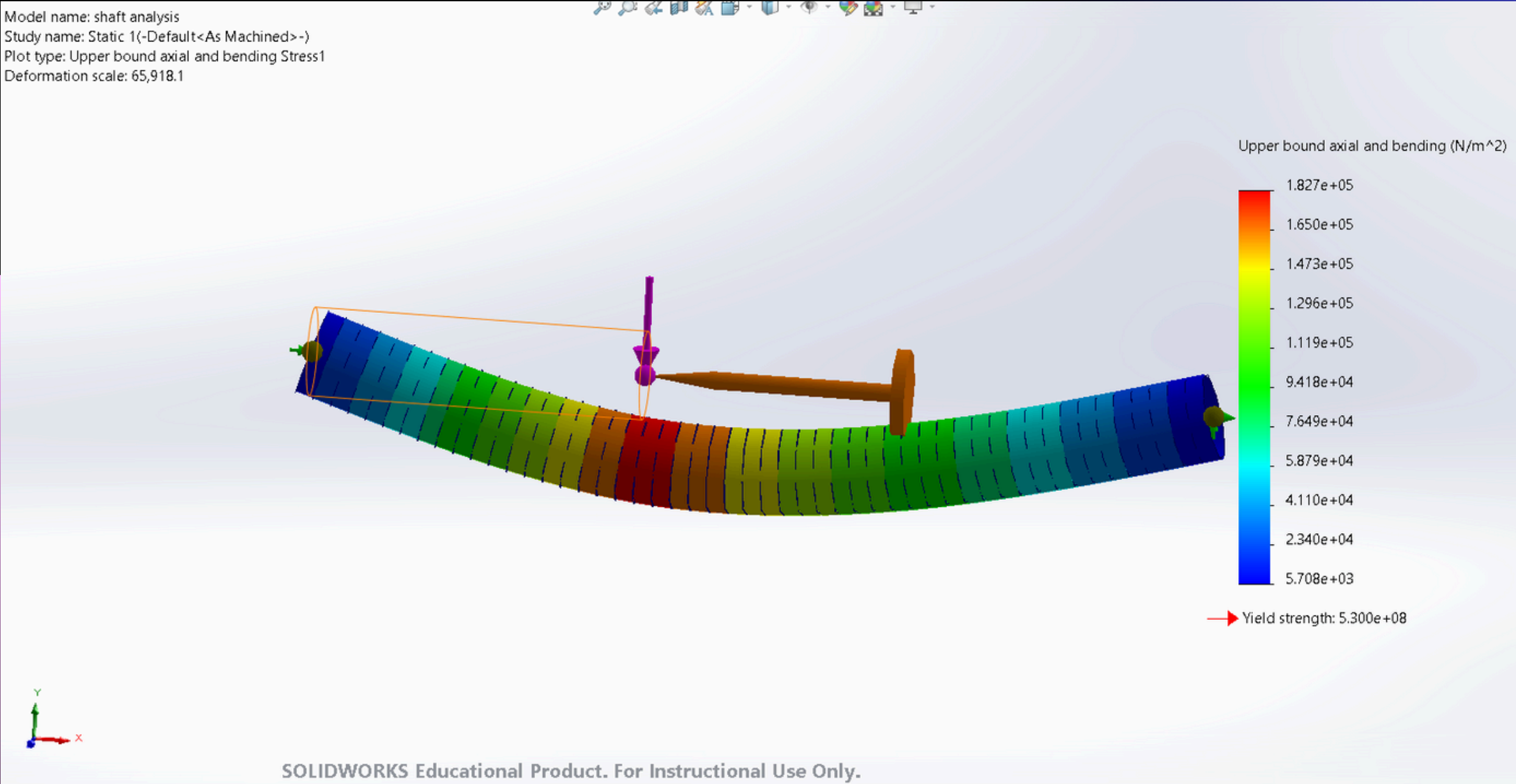


- Motor-driven vibrating sand sieving machine, designed to separate fine particles from coarse materials using a mechanically induced vibration mechanism.

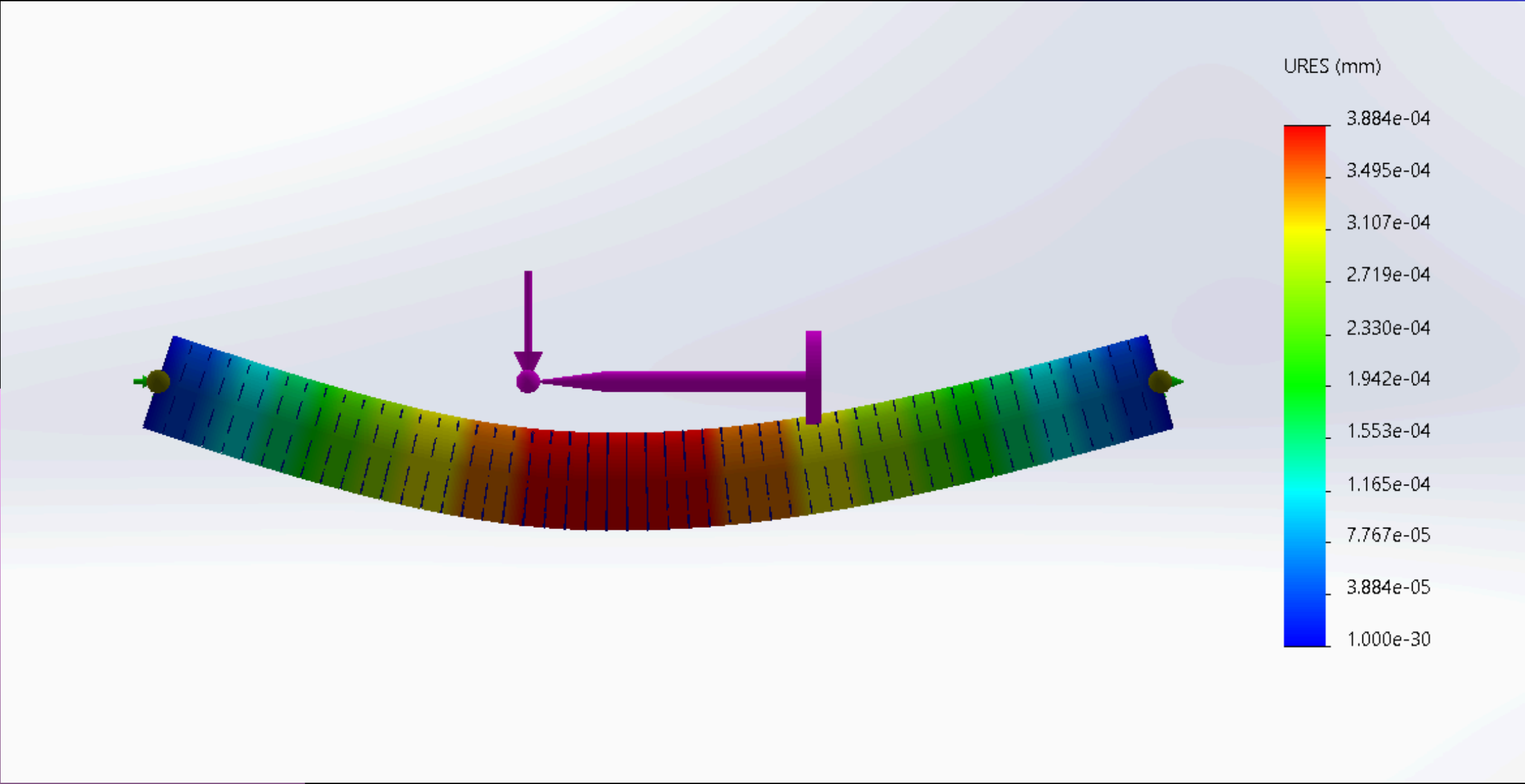
Motion Study



STRESS ANALYSIS



STRESS ANALYSIS



RESULTS AND CONCLUSIONS

- The proposed motor-driven crank–slider vibrating sieve reliably converts rotary motion into controlled reciprocating motion and meets the project objective of mechanical sieving.
- Material choices provide a good balance of strength, manufacturability and cost, while the stainless mesh ensures abrasion and corrosion resistance from sand.
- The frame and bearing selections ensure stable support and reduced vibration transmission, improving reliability and operator safety.
- Overall, the design is cost-effective, manufacturable, and suitable for small-scale agricultural or construction sieving applications, meeting the project requirements for stability, ease of use and continuous operation.

Sources

- **Design and Fabrication of Industrial Sand Screening Machine for Green Sand Soham**
V. Hapsenkar1* , Rahul M. Sasamkar2 , Ambika V. Kamatgikar3 , Sanket D. Mandlik4 ,
- **Development of a Mechanical Sand Sieving Machine with an Automatic Vibrator**
Ajayi, A. Bola1* & Oyeniya, R. Ayodeji2

Thank You